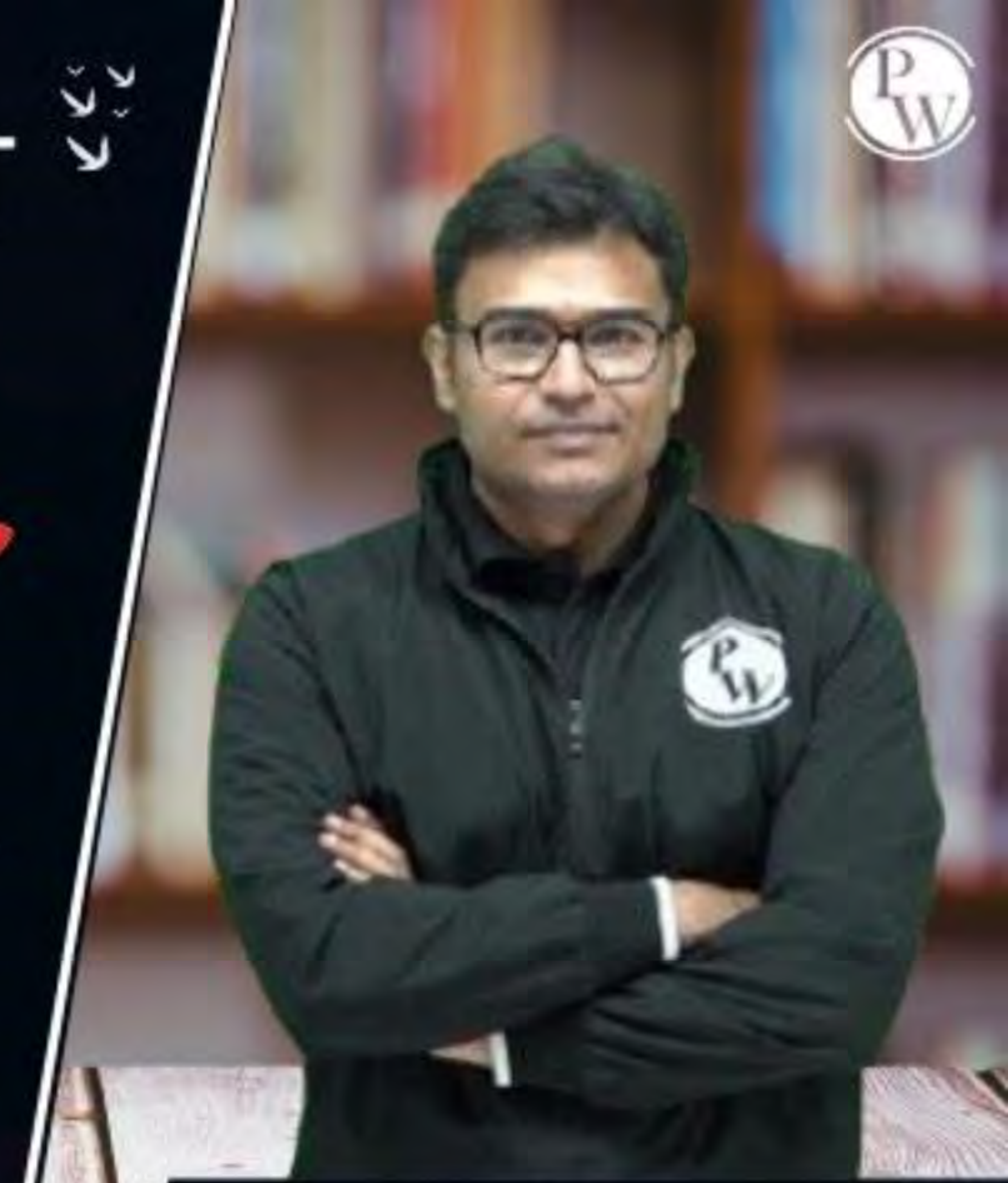
# Computer Science & IT

## Theory of Computation

Regular Languages

**Lecture No.** 1



By- Ravindra Sir

# Topics to be Covered

Topic

Topic

Topic

Regular Languages

TOC

60% Regular languages — 5-6

30% Context free languages - 5-6

10% Turing machine - 1 mark

12

Books: Ullman → ~~X~~

Peter Linz → 20% ~~X~~

75 hours

Note — Bible — TOC

a) very basics → basics → medium → Hard

b) medium → hard

c) hard

TOC

M, P, C

CS

Symbol — a, b, 0, !, A, B, C, ... Z, a, b ... z, 

$\Sigma = \{a, b\} \rightarrow$ alphabet

$\Downarrow$

$S_1 = a$

$S_2 = b$

$S_3 = ab$

$S_4 = aabb$

⋮

Language :- Set of strings

strings $\downarrow$ finite

Language → finite, Infinite

$\Sigma = \{a, b\}$

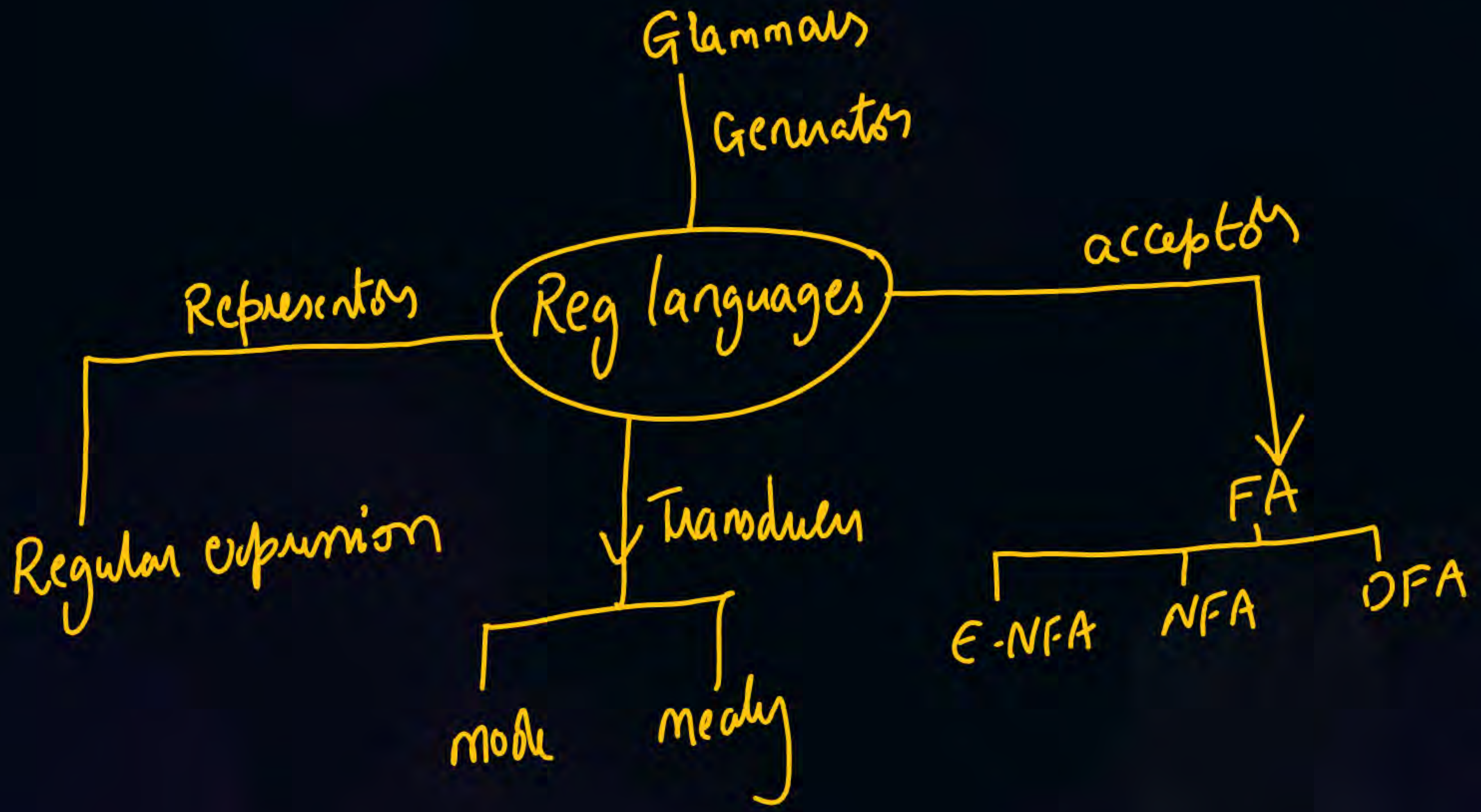
$L = \{\underline{ab, ba}\}$

$L_1 = \{$set of all strings starting with 'a'$\}$

$= \{a, ab, aab, aaa, \ldots\ldots\}$

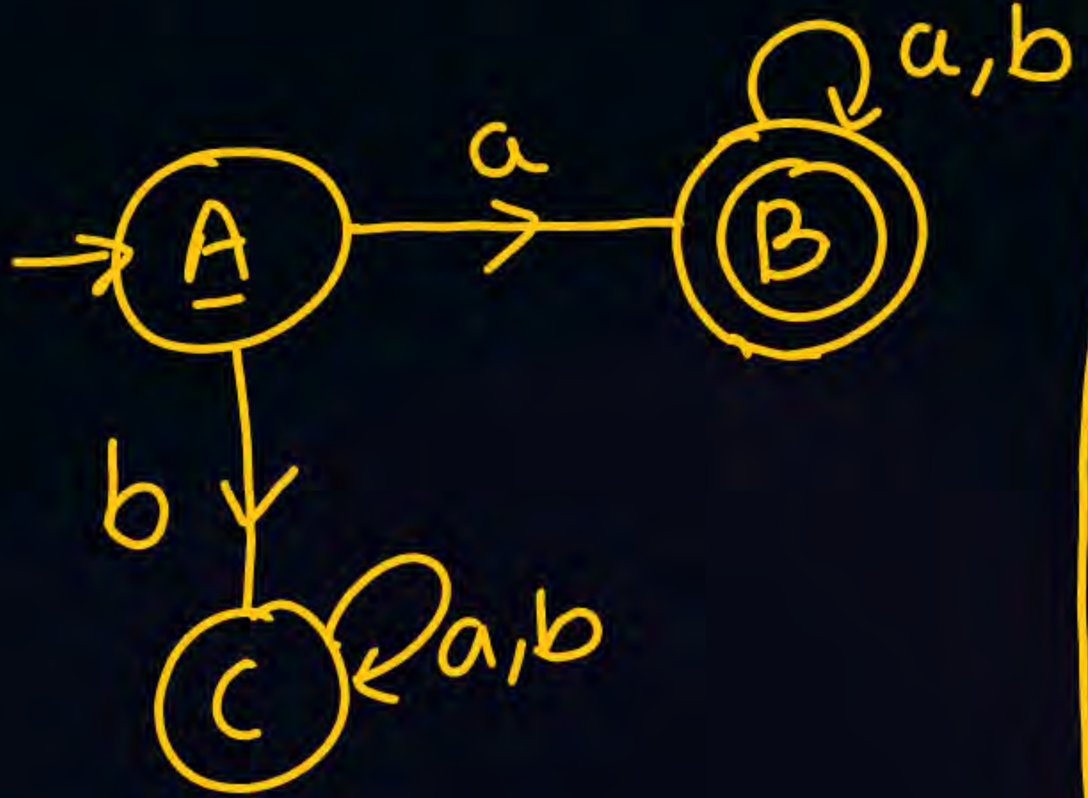
DFA →

Very basics


Grammars
Generator
Reg languages
Representor
Regular expression
Transducer
moore
mealy
acceptor
FA
ε-NFA
NFA
DFA

(Deterministic FA)

DFA : $(Q, \Sigma, \delta, q_0, F)$




$Q$ = finite set of states

$\{A, B, C\}$

$\Sigma = \{a, b\}$ = finite set of symbols

$\delta$ : Transition function

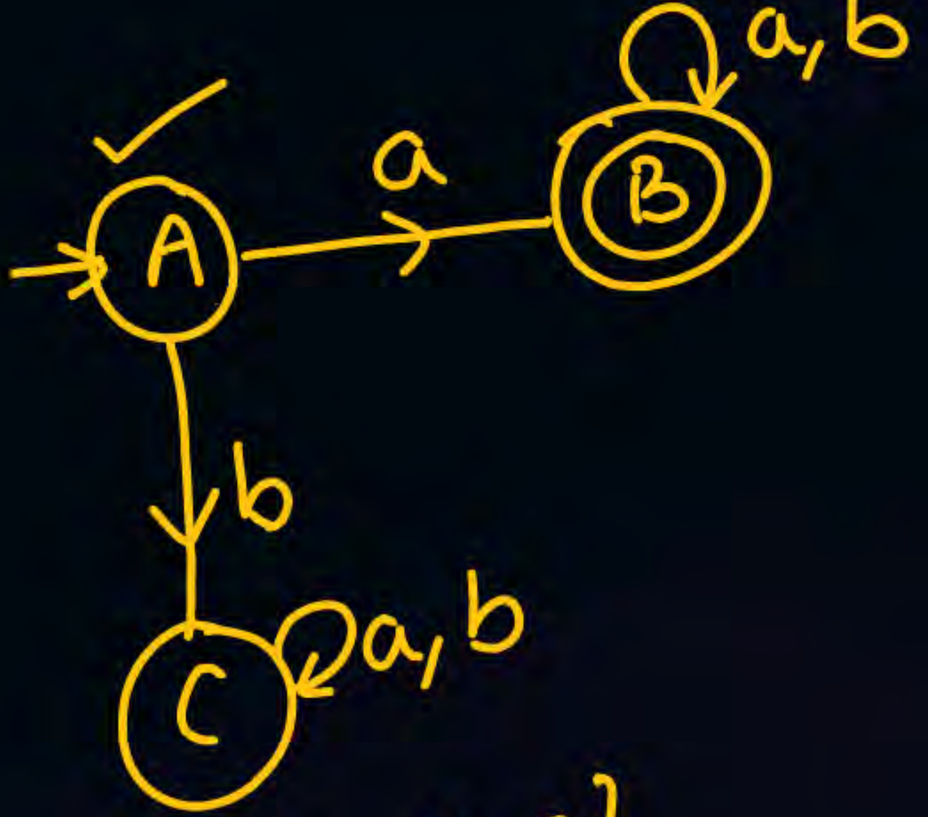
$$Q \times \Sigma \longrightarrow Q$$

$q_0 \rightarrow$ initial

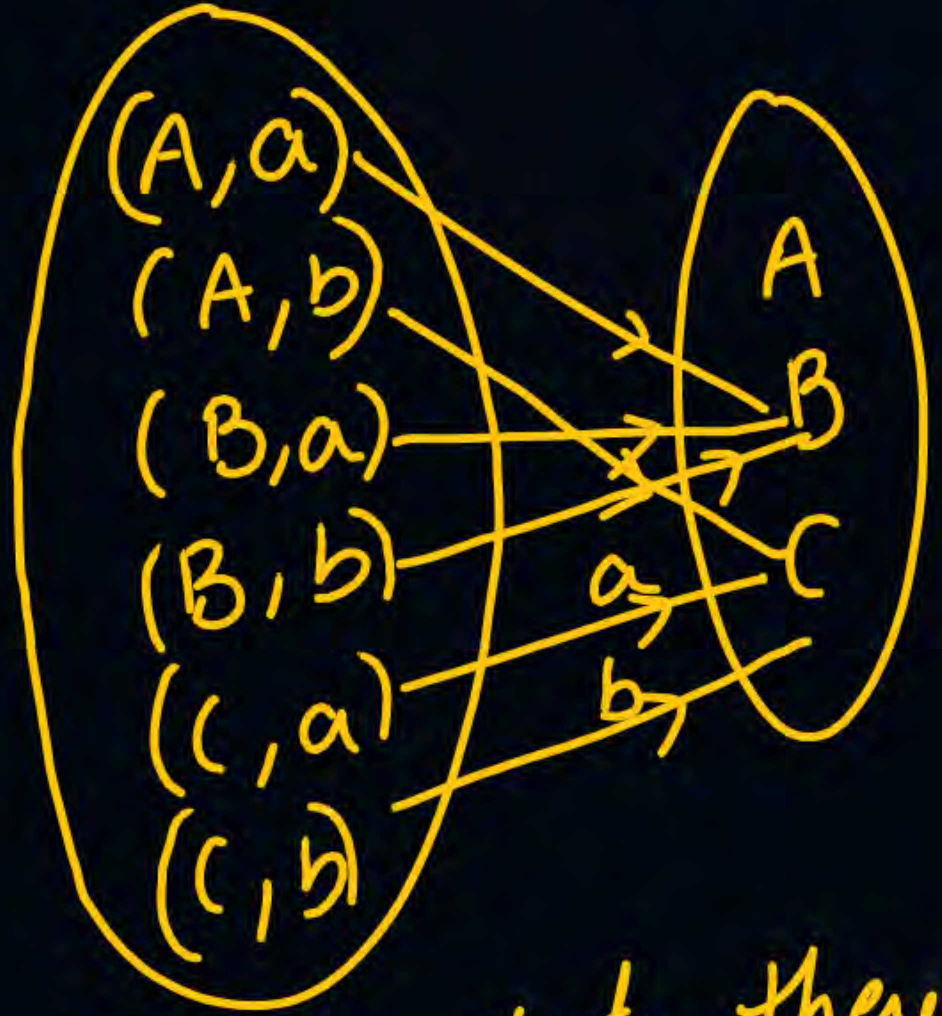
$q_0 = A$

$F$ = finite set of final states

$= \{B\}$




$\delta: Q \times \Sigma \rightarrow Q$



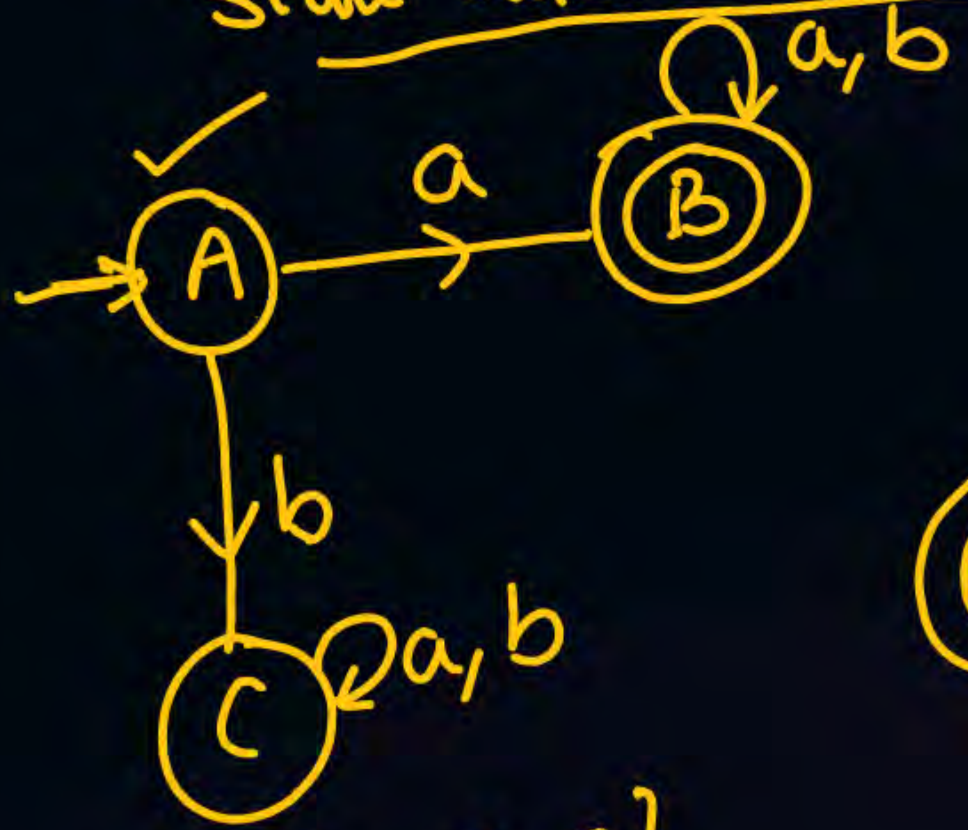

$Q = \{A, B, C\}$

$\Sigma = \{a, b\}$

$Q \times \Sigma = \{(A,a)\ (A,b)\ (B,a)\ (B,b)\ (C,a)\ (C,b)\}$

DFA:- For every state, there will one transition on every symbol

## State transition diagm




$Q = \{A, B, C\}$

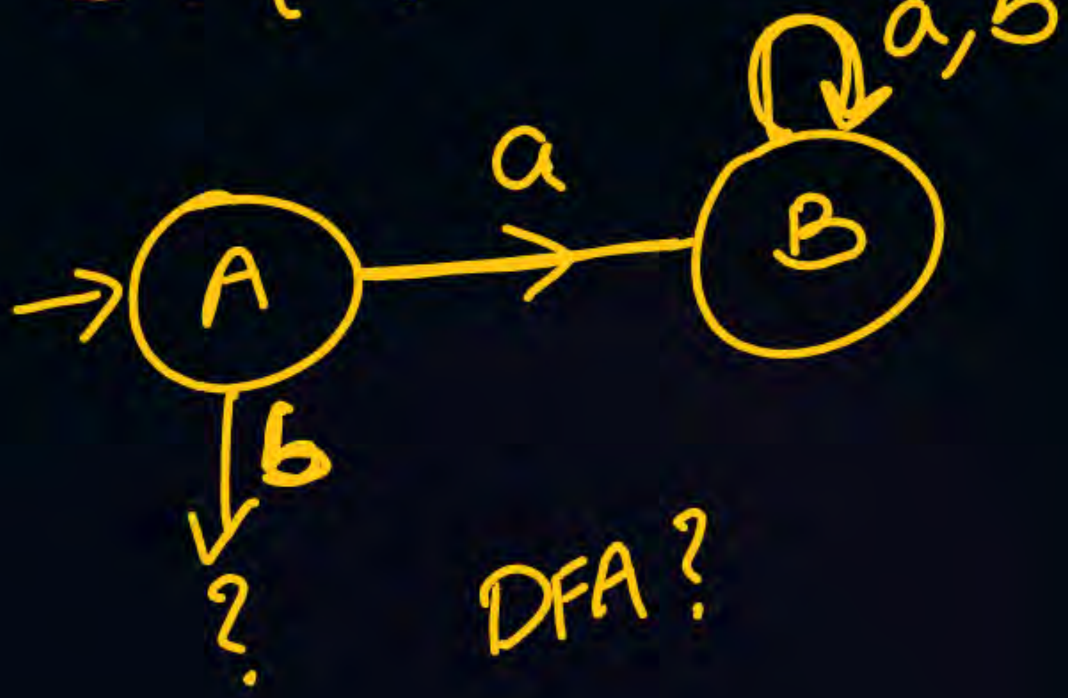
$\Sigma = \{a, b\}$

$Q \times \Sigma = \{(A,a)\,(A,b)$
$(B,a)\,(B,b)$
$(C,a)\,(C,b)\}$

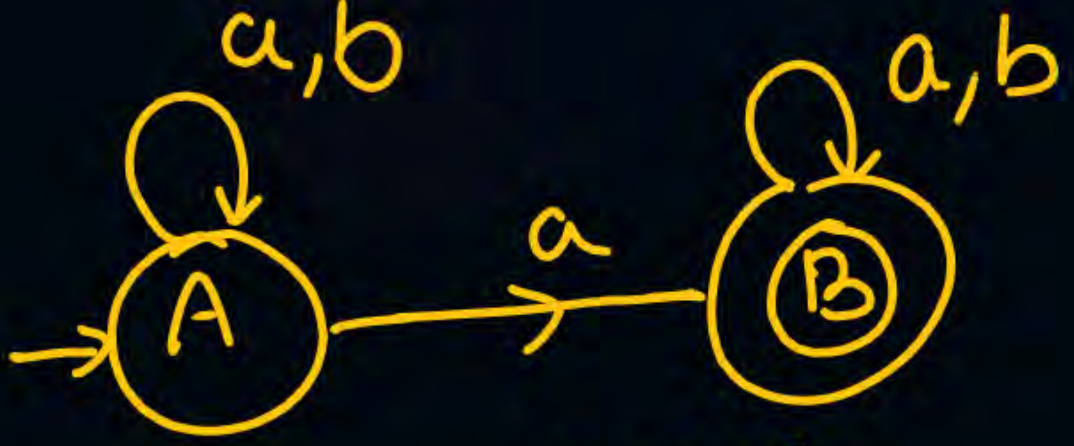
## State transition table

| | a | b |
|---|---|---|
| →A | B | C |
| *B | B | B |
| C | C | C |

$\Sigma = \{a, b\}$




DFA ?




DFA = ?

$A \xrightarrow{a} A$ ✓

$A \xrightarrow{a} B$ ✓

# Non deterministic FA :- NFA

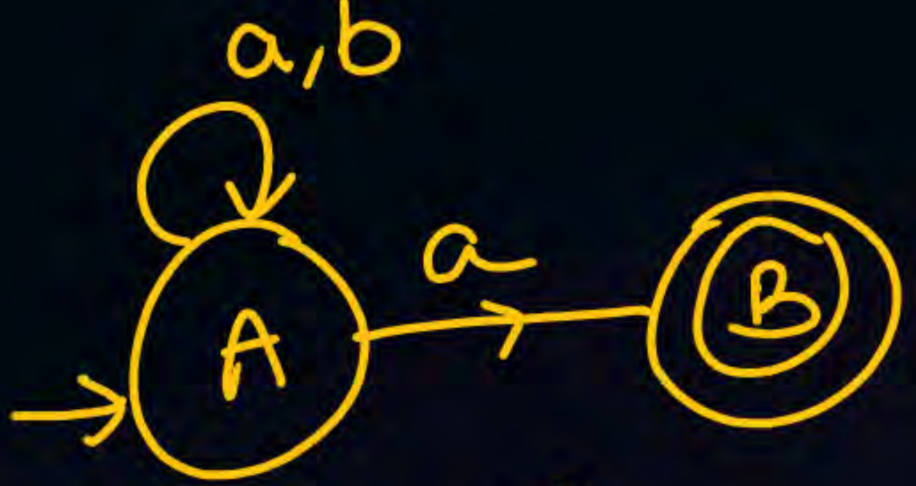
$(Q, \Sigma, \delta, q_0, F)$

$\delta : Q \times \Sigma \rightarrow 2^Q$

$Q = \{A, B\}$

$2^Q = \{ \{\}, \{A\}, \{B\}, \{AB\} \}$

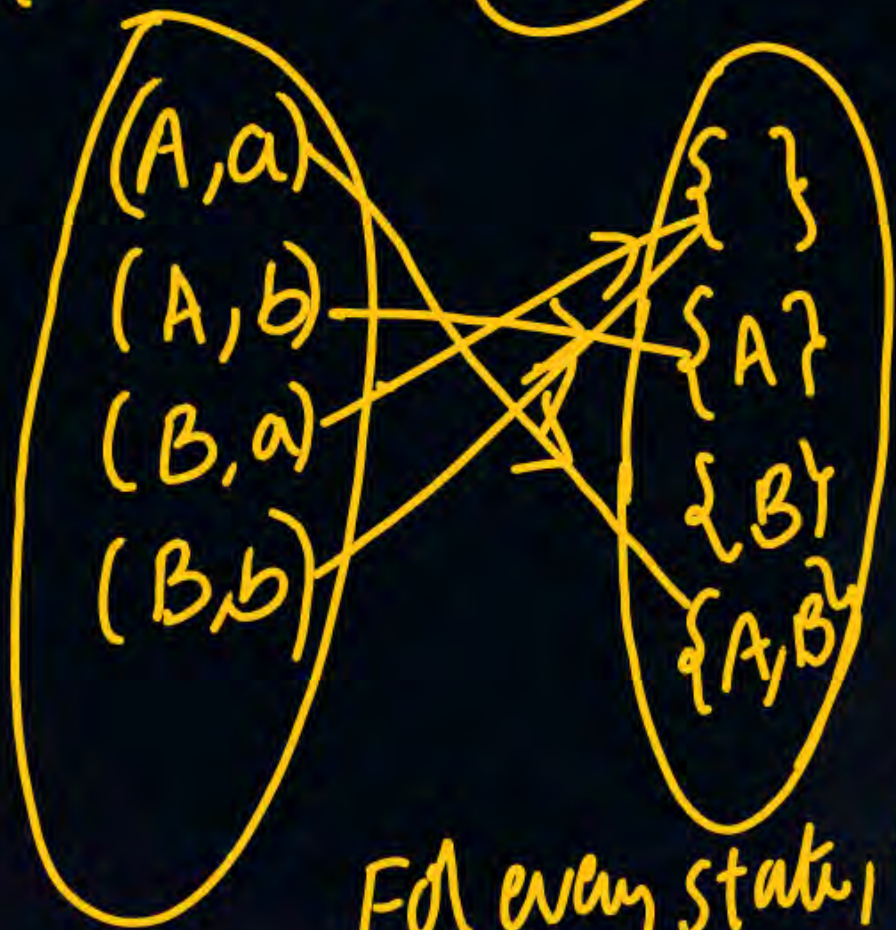
| | a | b |
|---|---|---|
| →A | A,B | A |
| *B | — | — |




$Q = \{A, B\}$

$\Sigma = \{a, b\}$

$Q \times \Sigma = \{(A,a)\,(A,b)\,(B,a)\,(B,b)$




For every state, for every symbol, we many get zero or one or more transition


$Q = \{A, B\}$

**NFA** ✓

$\delta : Q \times \Sigma \rightarrow 2^Q$

$\{\ \}$

$\{A\}$

$\{B\}$

$\{A, B\}$

**DFA** ✓

$\delta : Q \times \Sigma \rightarrow Q$

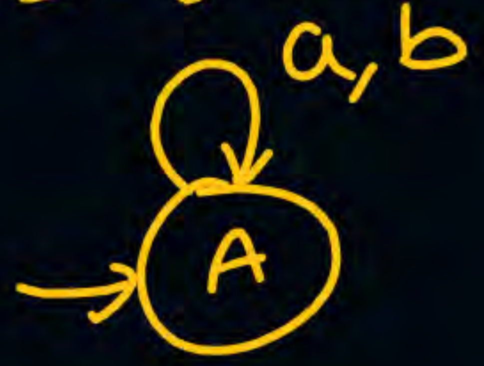
A

B

Every DFA is an NFA.

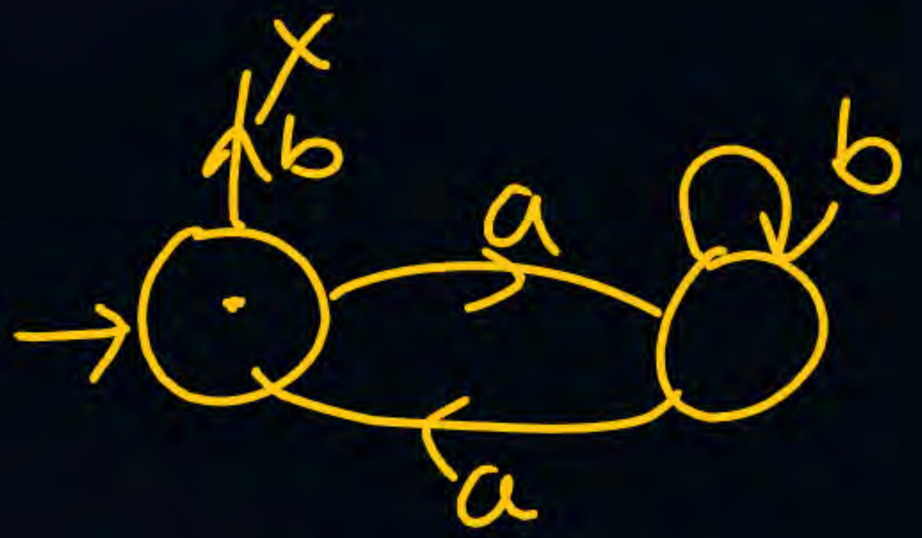
Power(NFA) = Power(DFA) < Power(RBR) ✓

$\Sigma = \{a, b\}$




DFA

NFA




NFA

✗ DFA

# Reg expressions :-

## Operators :-

$\oplus \longrightarrow$ union

$\odot \longrightarrow$ concat

$\circledast \longrightarrow$ Kleene closure

$a+b$

$a \cup b$

$a \cdot b = ab$

$a^{*} = a^0, a^1, a^2, a^3, \ldots, a^n, \ldots$

$= \epsilon, a, aa, aaa \ldots$

a) $\phi$, $\epsilon$, $(a, b) \in \Sigma$

b) $r_1, r_2 \rightarrow$ Reg expression,

$r_1 + r_2,\ r_1 \cdot r_2,\ r_1^*$ RE

$RE = \epsilon$

$L = \{\epsilon\}$

$\Sigma = \{a, b\}$

$\Sigma^2 = \{a, b\} \cdot \{a, b\} = \{aa, ab, ba, bb\} \rightarrow$ length 2

$\Sigma^3 = \{a, b\} \cdot \{a, b\}\{a, b\} = \{aaa \ldots\ldots bbb\}$ - length 3

$\Sigma^{*}$ 0 1 2 3 4 ....

$\Sigma^n$ = Set of all strings of length 'n'

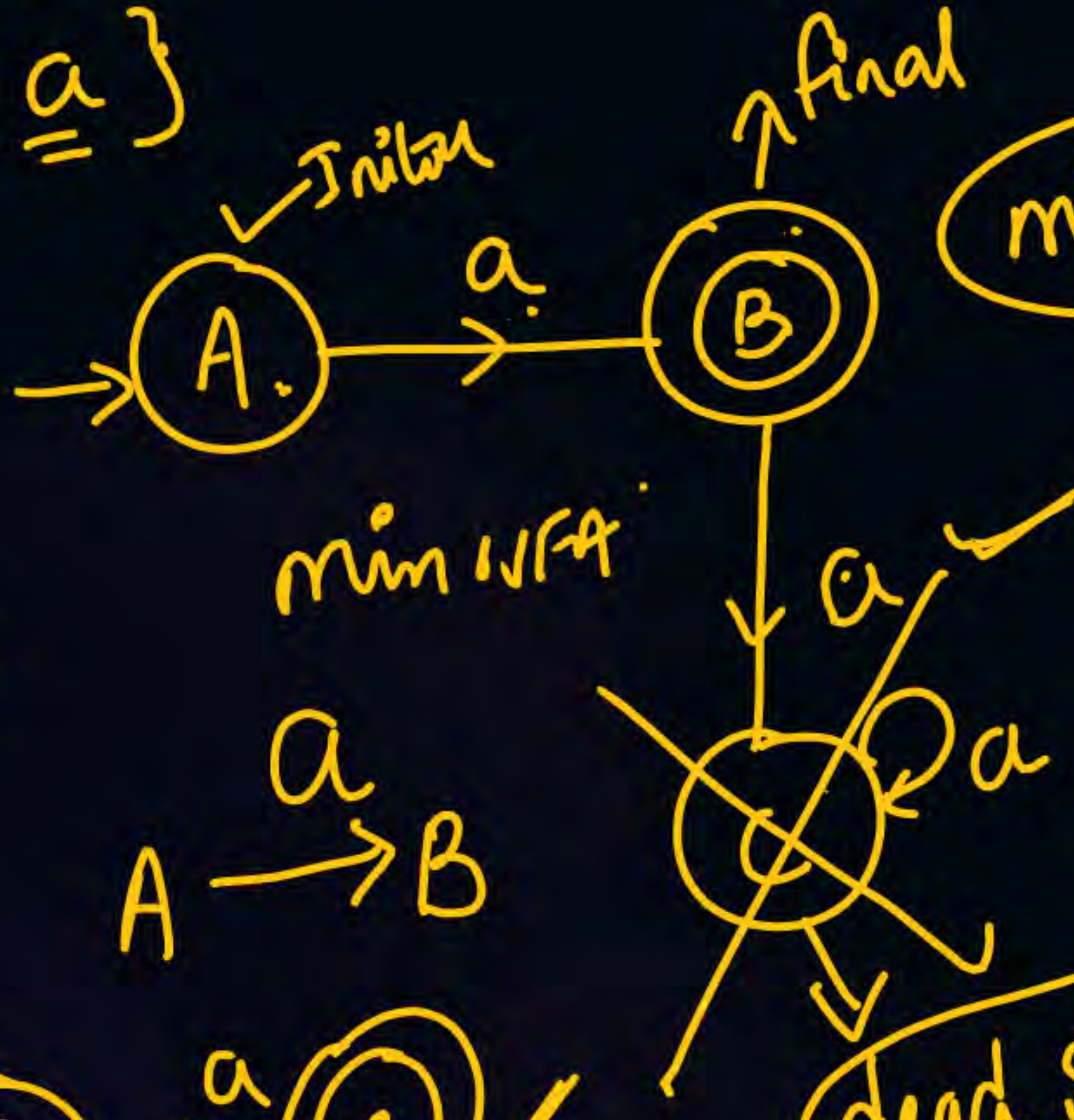
$\Sigma^0$ = String of length '0' = '$\epsilon$'   $|\epsilon| = 0$

$\Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \ldots \Sigma^n \ldots = \Sigma^{*} \rightarrow$ universal lang

Draw a DFA that accepts the language $\{a\}$

$\Sigma = \{a\}$

$L = \{a\}$



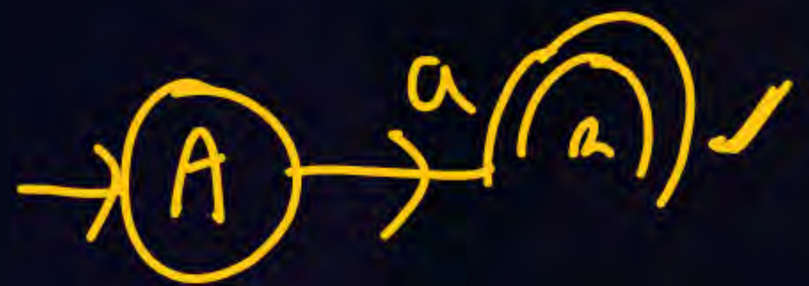

min DFA

$L = \{a\}$

$RE = \{a\}$

min NFA

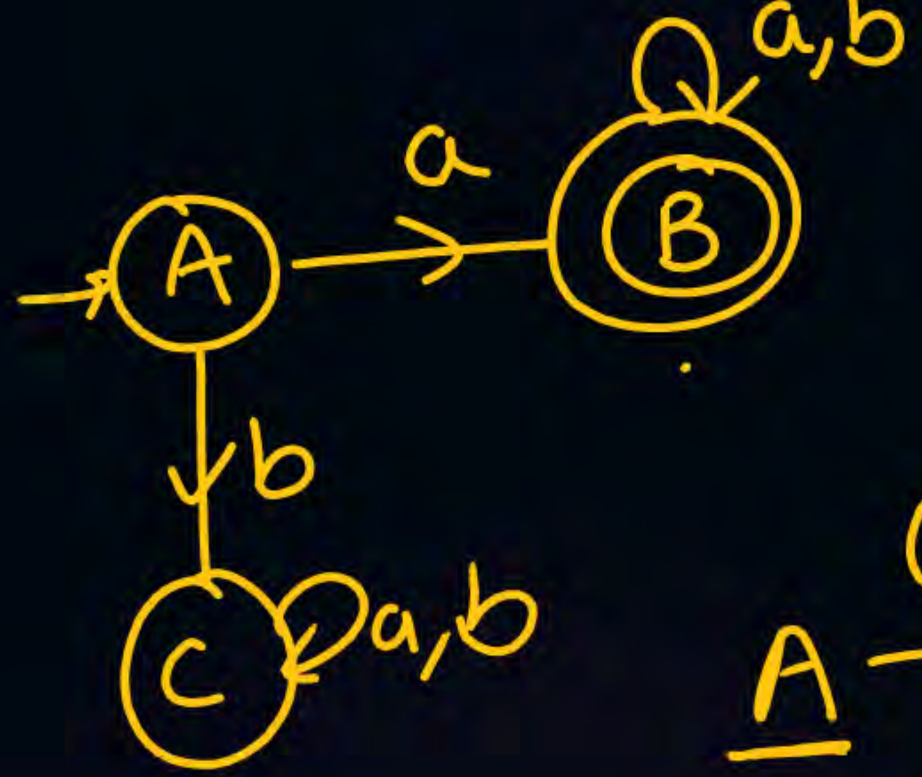
$A \xrightarrow{a} B$




$A \xrightarrow{a} B \xrightarrow{a} C$ X

dead state.

A state from which we cannot reach final state.



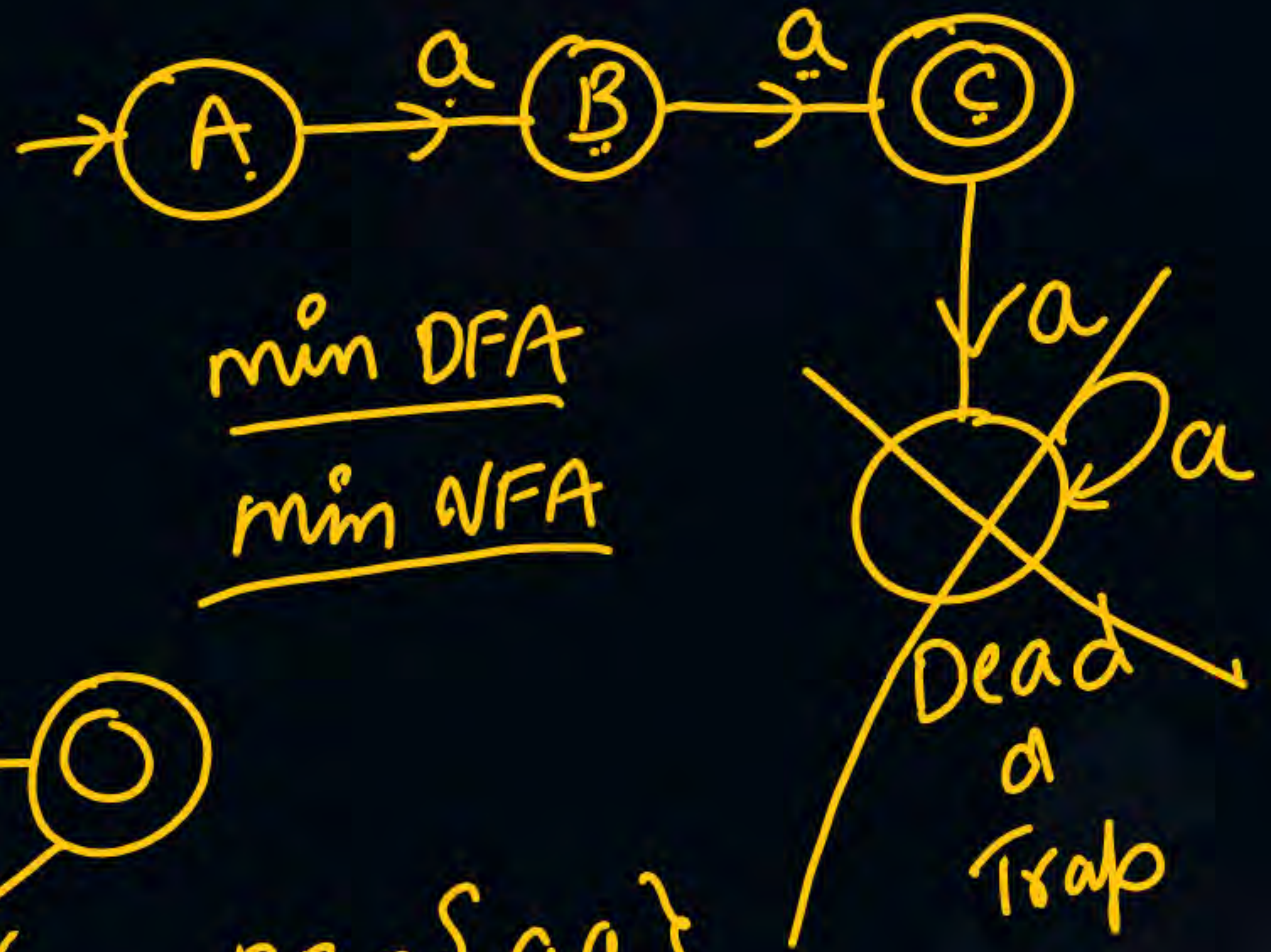

$$\underline{A} \xrightarrow{a} B \xrightarrow{a} B \xrightarrow{b} B \xrightarrow{b} B \checkmark$$

$$\underline{\underline{A}} \xrightarrow{b} C \xrightarrow{a} C \xrightarrow{a} C \xrightarrow{b} C \times$$

$\Sigma = \{a\}$ min DFA

$L = \{aa\}$ min NFA

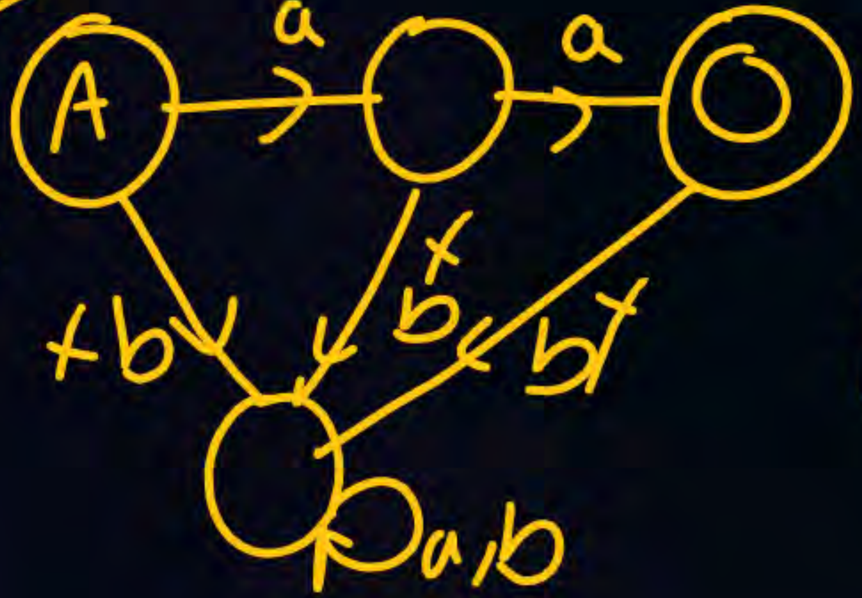
RE




min DFA

min NFA


aa; aaaaa

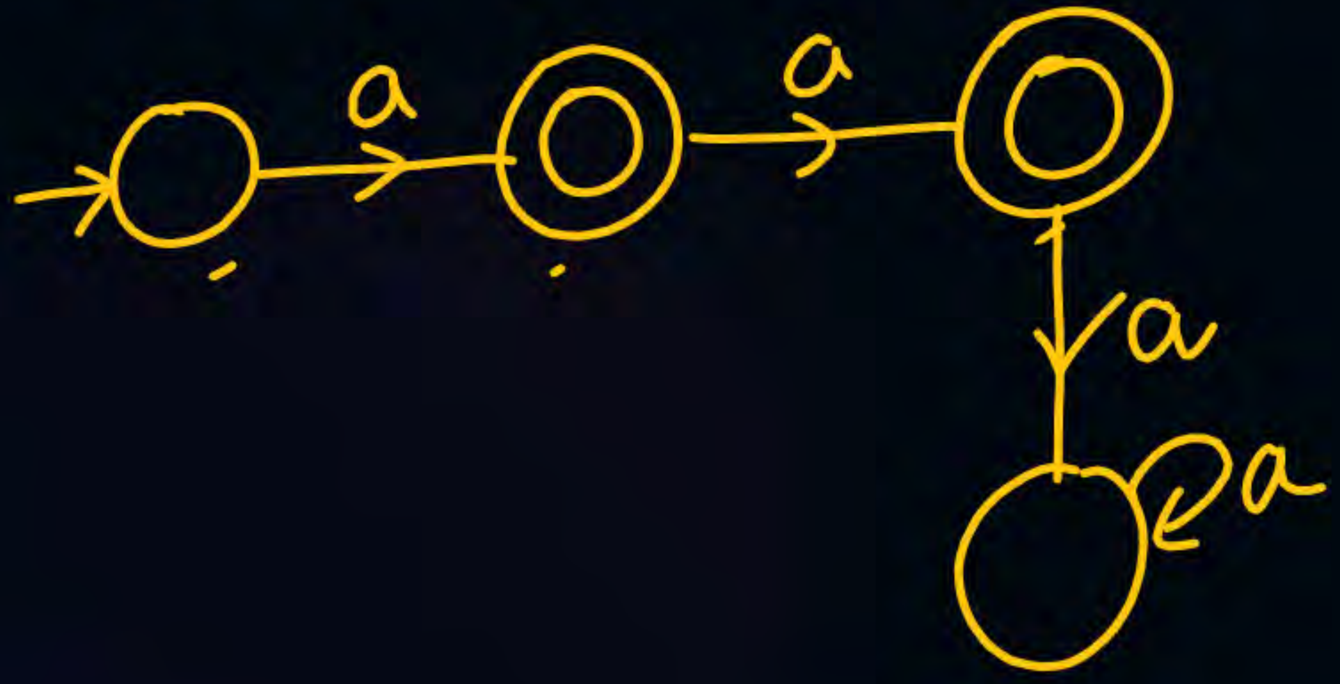



RE = {aa}

$\Sigma = \{a\}$

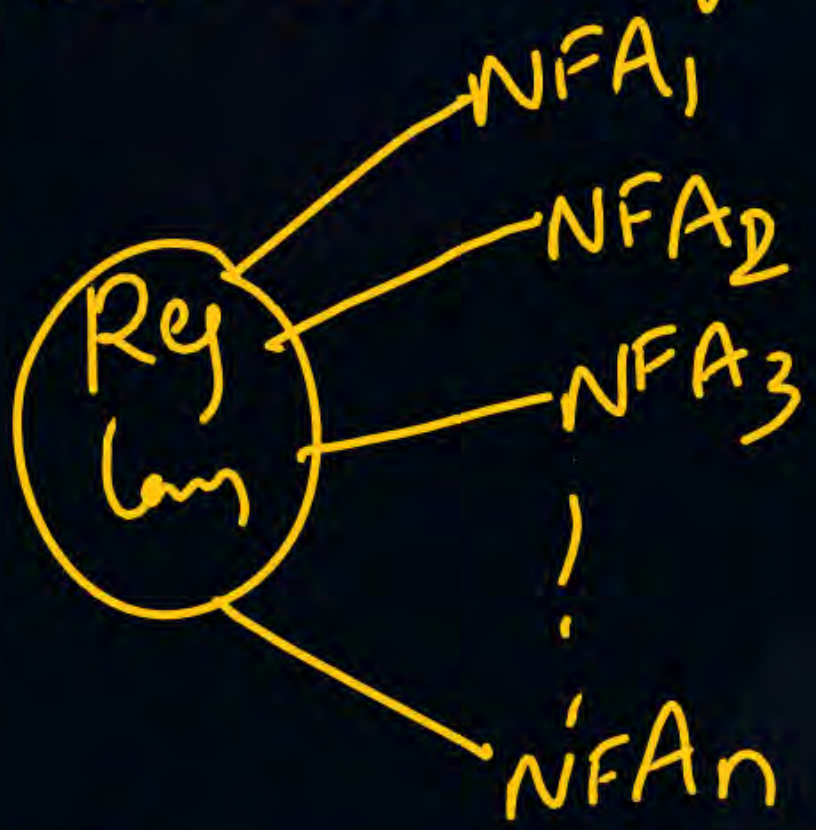
$L = \{a, aa\}$

DFA
NFA
RE



$\Sigma = \{a\}$ L = Set of all strings with even no of a's


min DFA is always unique



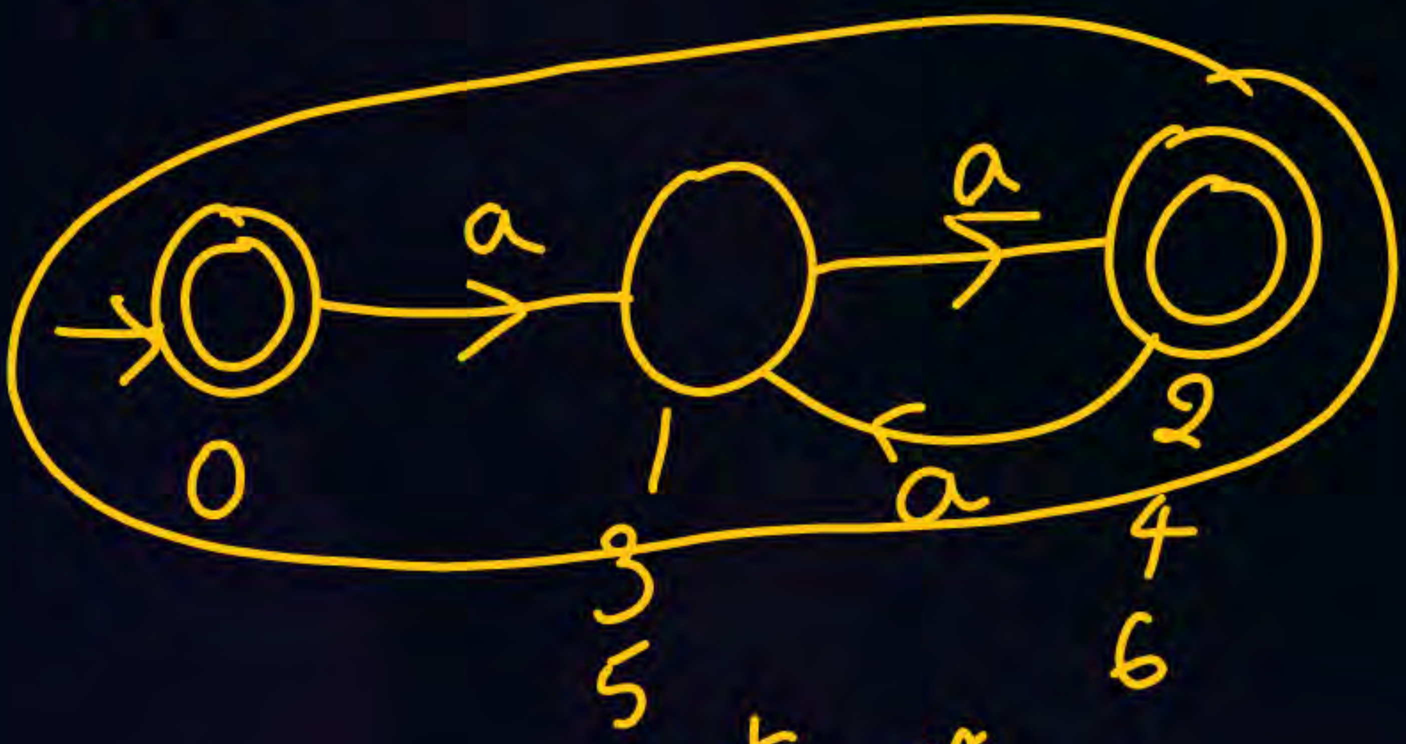

min NFA is not unique.

$\Sigma = \{a\}$ even no of a's

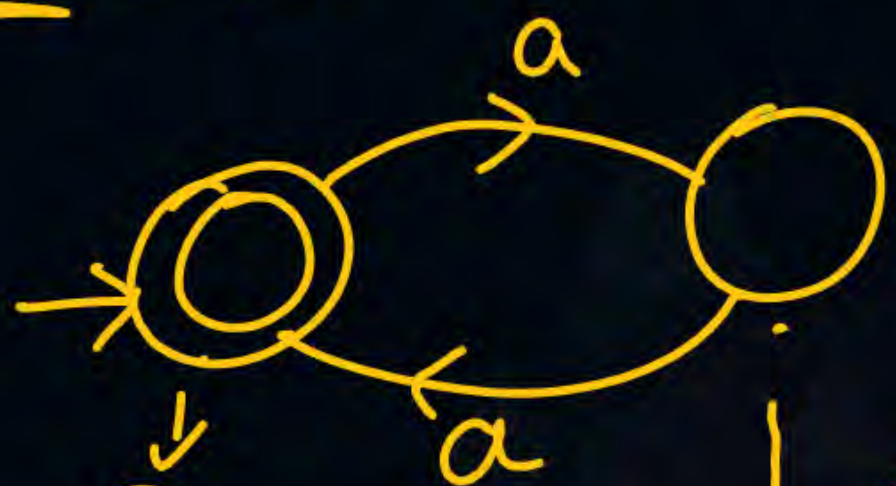
$L = \{\epsilon, aa, aaaa, \ldots\}$

If a NFA or a DFA has to accept '$\epsilon$' then initial state should also be final

$RE = (aa)^* = \{\epsilon, aa, aaaa, \ldots\}$



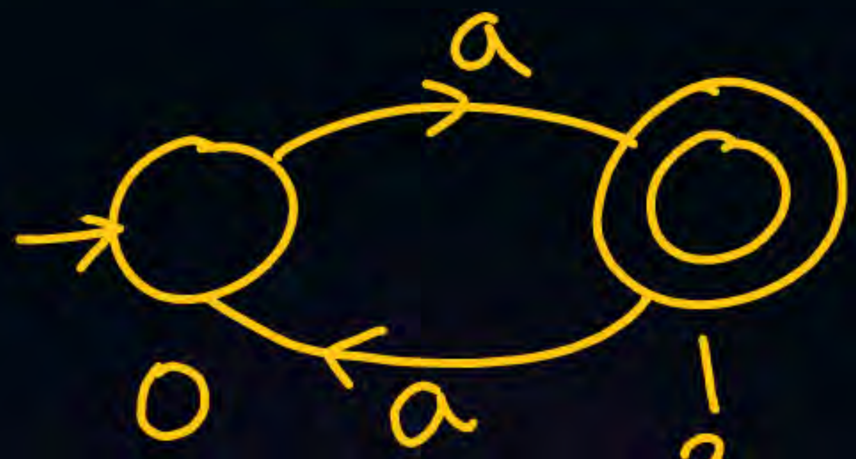

not min DFA



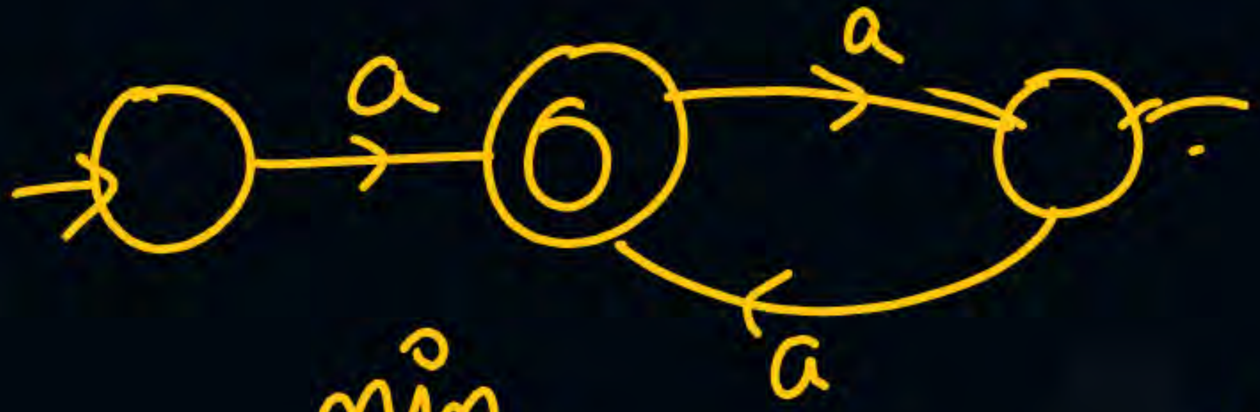

$\epsilon$
aa
aaaa
even ✓

odd ✓

$\Sigma = \{a\}$ $L = \{$ set of all string with odd no of a's $\}$




| | | |
|---|---|---|
| 0 | | 1 |
| 2 | min | 3 |
| 4 | DFA | 5 |
| 6 | min | 7 |
| ⋮ | DFA | ⋮ |

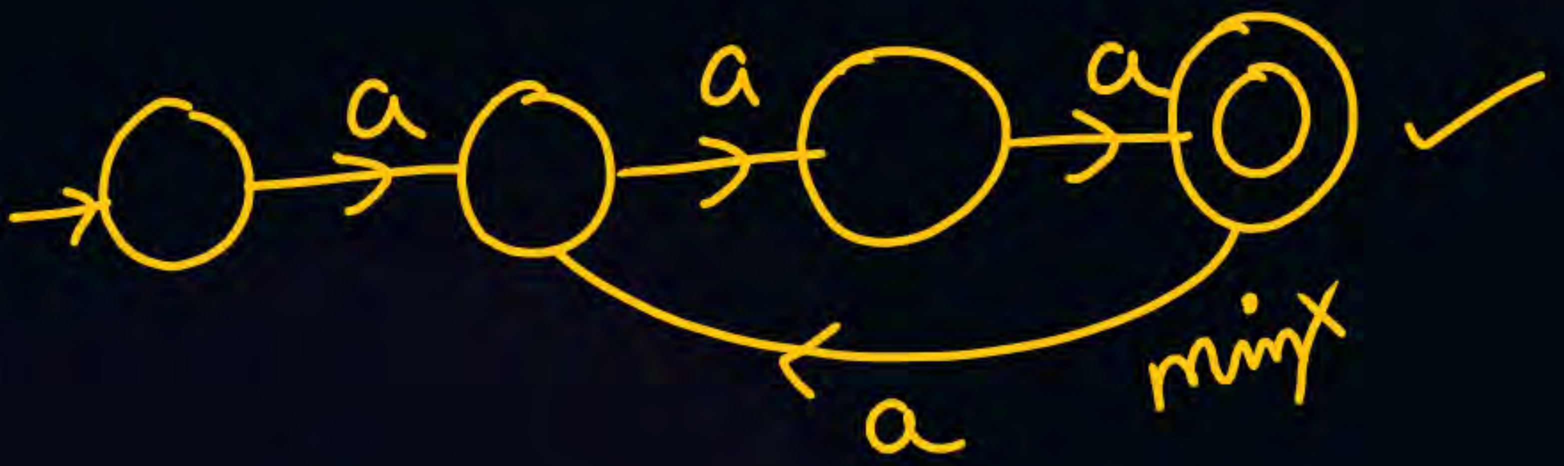


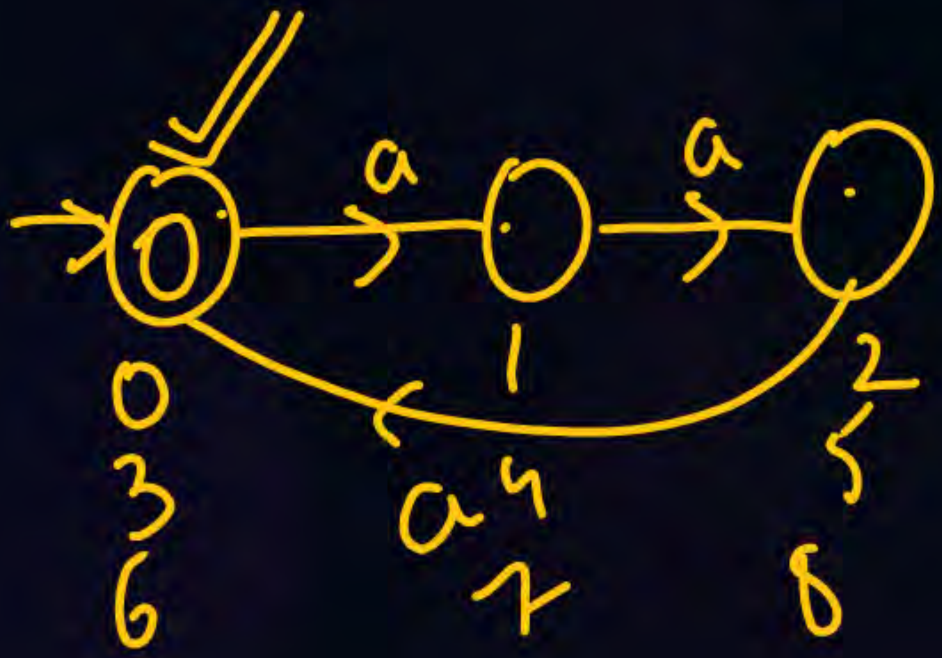

min

RE = $(aa)^*a$ & $a(aa)^*$

$\Sigma = \{a\}$ L = Set of all strings whose length is multiple of 3

$L = \{\epsilon, aaa, a^6, a^9 \ldots\}$

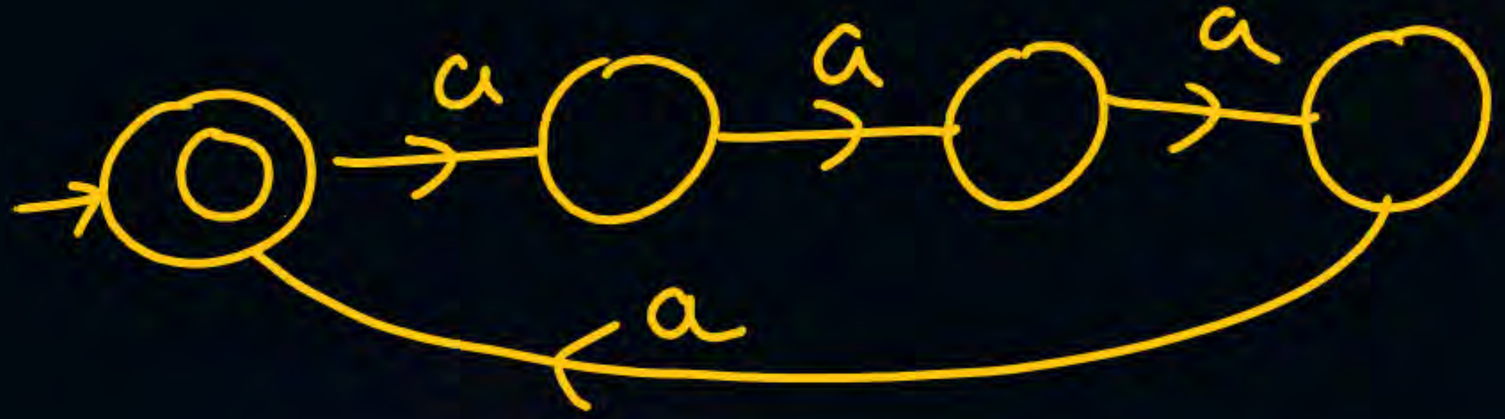



min DFA

min NFA

RE = $(aaa)^*$



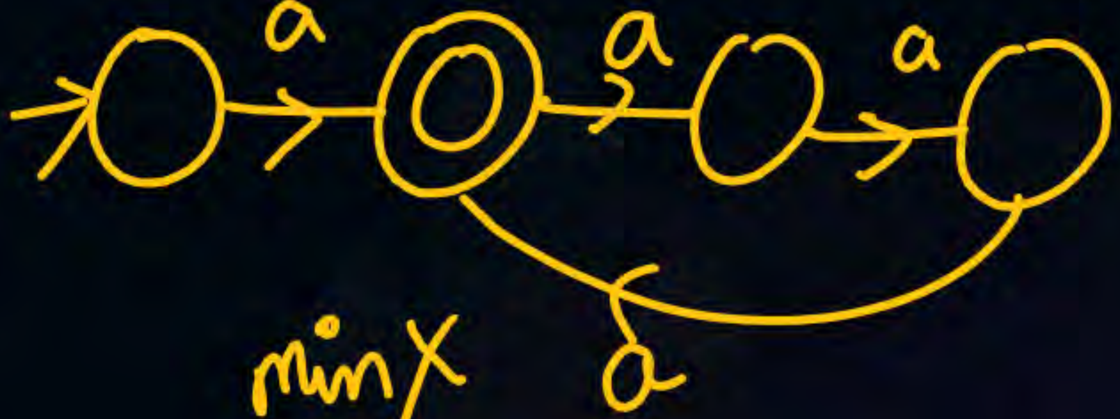

multiple of 100 → 100 states

$\Sigma = \{a\} \quad L = \{ w \,/\, n_a(w) \cong 1 \bmod 3 \}$


min DFA

min NFA

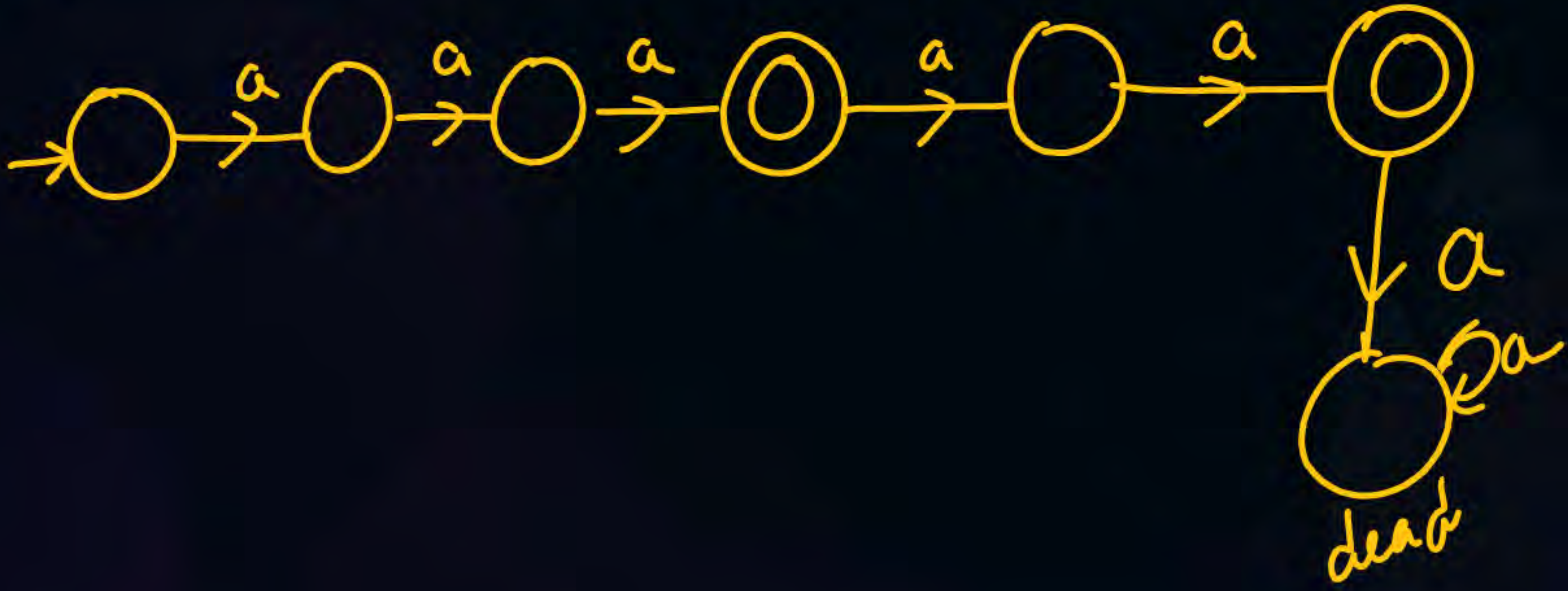
RE = $\boxed{(aaa)^* a}$




~~min~~

$\{a^n / n = 3 \text{ or } 5\}$

$L = \{aaa, aaaaa\}$



$L = (aa + aaa)^*$ RE$_g$

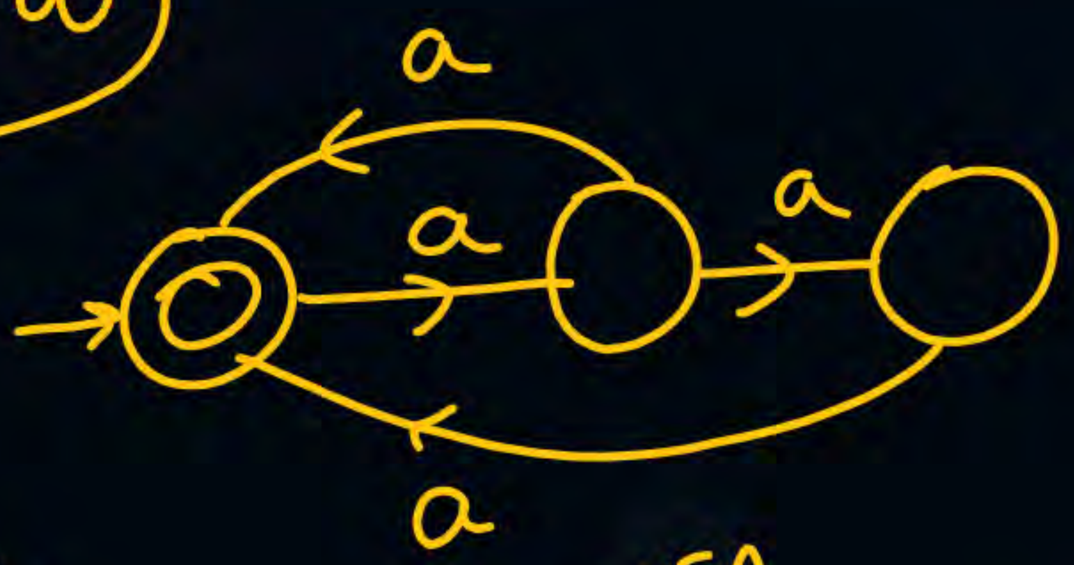
RBR Sir PW

min DFA

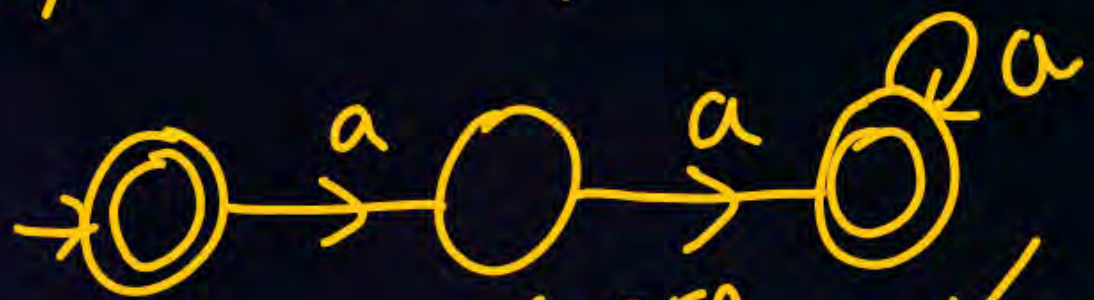
min NFA

$(aa + aaa)^*$

$\{\epsilon, aa, aaa, aaaa, aaaaa, \ldots\}$




min NFA




min DFA
min NFA

min NFA need not be unique

There can be more than one min NFA

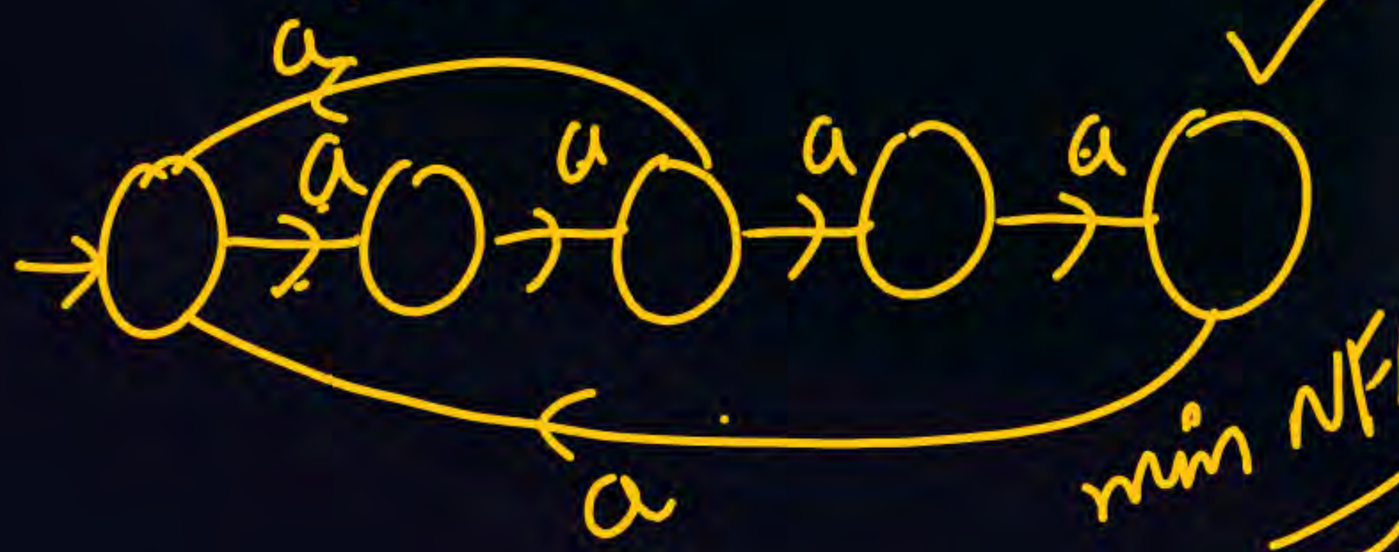
$(aaa + aaaaa)^{*}$ 5 min

min DFA

min NFA

$L = \{\epsilon, a^3, a^5, a^6, a^8, a^9, a^{10}, a^{11}, a^{12} \ldots\ldots\}$

min DFA

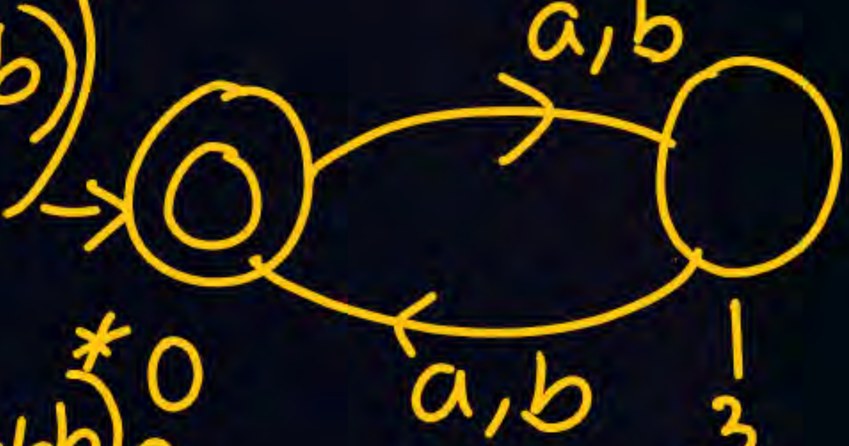


min NFA

$\Sigma = \{a, b\}$  $L = \{ w / |w| = \text{even} \}$

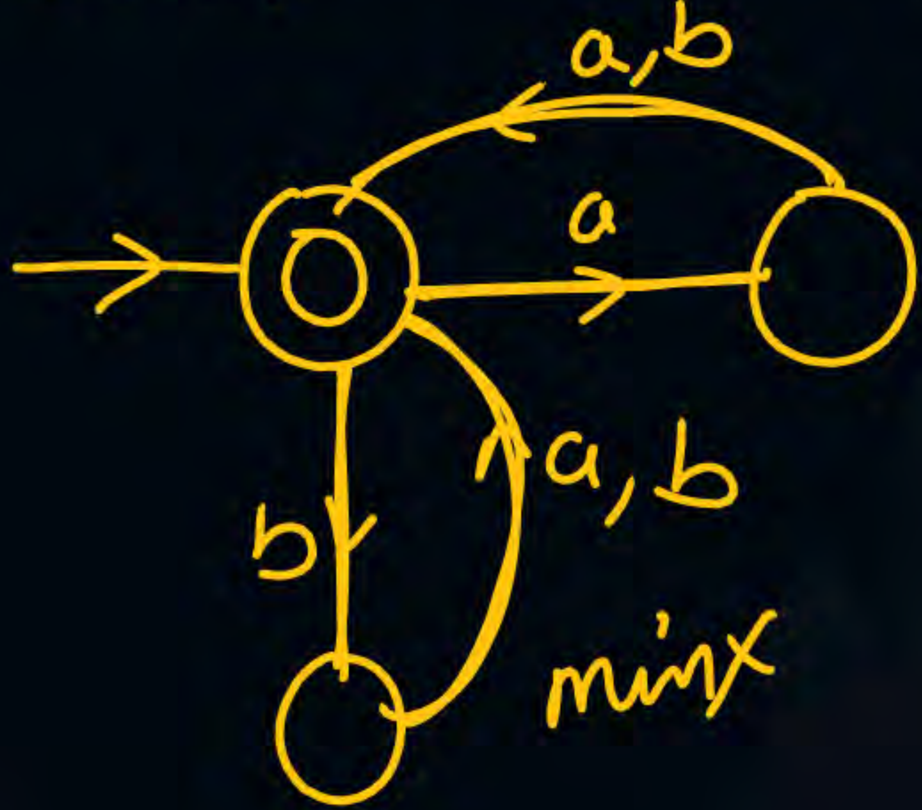
$L = \{ \epsilon, aa, bb, ba, ab, aaaa, aabb \ldots\ldots \}$

$((a+b)(a+b))^*$




$(aa+ba+ab+bb)^*$

| | | |
|---|---|---|
| 0 | | 1 |
| 2 | | 3 |
| 4 | min DFA | 5 |
| 6 | | 7 |
| . | min NFA | . |
| . | | . |




min ✗

$\Sigma = \{a,b\}$ $|w| = \text{odd}$

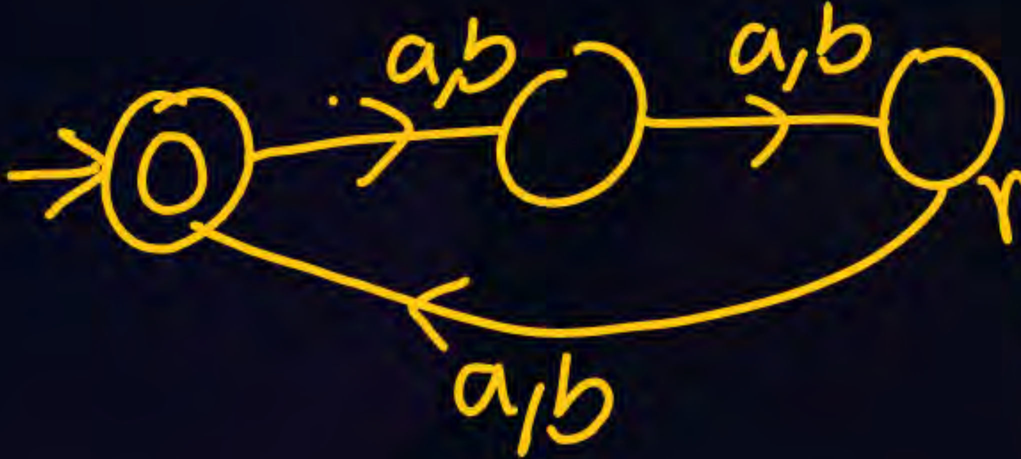

min ✗

✓


$\big((a+b)(a+b)\big)^*(a+b)$ ✓

$\cancel{(a+b)^{2n}(a+b)}$

$\cancel{(a+b)^{2n+1}}$




min DFA

$\Sigma = \{a, b\} \quad |w| = 3n \,/\, n \geq 0$

$((a+b)(a+b)(a+b))^*$

~~Saturday~~ ✗

Sunday ✓

NOTES — Telegram channel

Easy → medium → Hard

# THANK - YOU

ultimate TOC course ✓
↓
Advanced TOC course ✓
↓
TOC marathon ✓